

Low Power Consumption

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

This document describes low power consumption setting. This setting includes vertical blanking low power consumption setting and horizontal blanking low power consumption setting. The vertical blanking low power consumption setting reduces power consumption by stopping the sensor internal circuit during vertical blanking period. The horizontal blanking low power consumption setting reduces power consumption by stopping the sensor internal circuit during horizontal blanking period of each line. In this sensor, both the vertical blanking low power consumption setting and the horizontal blanking low power consumption setting are always ON.

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3. Features

◆ Low Power Consumption mode

Horizontal blanking Low Power Consumption mode
Vertical blanking Low Power Consumption mode

◆ Support mode

Normal mode (XVS / XHS sync)
Sequential Trigger mode
Fast Trigger mode

◆ Support readout drive mode

All-pixel scan mode
Vertical / Horizontal 1/2 Subsampling mode
2 × 2 FD binning mode (The product with a color filter is excluded.)
ROI mode

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

Contents

1.	Product type -----	1
2.	Description -----	1
3.	Features -----	2
	Contents-----	3
4.	Register Map -----	4
5.	About Vertical Blanking Low Power Consumption Setting-----	8
5.1.	Overview -----	8
5.2.	List of Settings -----	8
6.	About Horizontal Blanking Low Power Consumption Setting -----	9
6.1.	Overview -----	9
6.2.	List of Settings -----	10
7.	Note-----	11
8.	Trouble case-----	12
	Revision History-----	14

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4. Register Map

The register map for low power consumption is shown below.
Please refer to the datasheet for register settings other than those listed.

Registers corresponding to Chip ID = 02h in Write mode. (I²C:30**h)

Address		bit	Register Name	Description	Default value after reset						Reflection timing
4-wire	I ² C				By register			By address			
D8h	30D8h	0	HMAX [15:0]	LSB	00A6h *2	0081h *3	0167h *4	A6h *2	81h *3	67h *3	S
		1									
		2									
		3									
		4									
		5									
		6									
		7									
D9h	30D9h	0		Horizontal span setting							
		1		Sensor master mode: Refer to the datasheet							
		2		Sensor slave mode:							
		3		HMAX = "Number of INCK in 1H" × 74.25							
		4		/ INCK Frequency							
		5									
		6									
		7	MSB								

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3400h	0	TRIGMODE [2:0]	Global shutter mode setting 0h: Normal mode 1h: Sequential trigger mode 2h: Fast Trigger mode Others: Setting prohibited	00h	00h	I ^{*1}
		1					
		2					
		3	TRIGTIMING [1:0]	Trigger mode setting 0h: Normal mode 1h: Trigger mode Others: Setting prohibited			I ^{*1}
		4					
		5		Fixed to 0			
		6	Fixed to 0	—			
		7	Fixed to 0	—			

*1 Refer to "Mode Transitions of Global Shutter Operation" in the datasheet.

*2 IMX530 series and IMX540 series.

*3 IMX535 series, IMX545 series and IMX565 series.

*4 IMX548 series and IMX568 series.

Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset				Reflection timing	
4-wire	I ² C				By register		By address			
9Ch	359Ch	0	VSST_EN [3:0]	Vertical blank low power consumption enable 0h: Disable, Fh: Enable	Fh ^{*2}	0h ^{*5}	0Fh ^{*2}	00h ^{*5}	S	
		1								
		2								
		3								
		4	Fixed to 0	0	—					
		5	Fixed to 0	0	—					
		6	Fixed to 0	0	—					
		7	Fixed to 0	0	—					
A4h	35A4h	0	HSST_START1 [11:0]	LSB Horizontal blank low power consumption setting 1 Stopping internal circuit start timing designated (Refer to “List of Settings” item.)	29Ah ^{*2}	FFFh ^{*5}	9Ah ^{*2}	FFh ^{*5}	S	
		1								
		2								
		3								
		4								
		5								
		6								
		7								
A5h	35A5h	0		MSB			12h ^{*2}	0Fh ^{*5}	S	
		1								
		2								
		3								
		4	HSST_EN1 [0]	Horizontal blank low power consumption setting 1 enable 0: Disable, 1: Enable	1 ^{*2}	0 ^{*5}			—	
		5		Fixed to 0	0	—				
		6		Fixed to 0	0	—				
		7		Fixed to 0	0	—				
A6h	35A6h	0	HSST_END1 [15:0]	LSB Horizontal blank low power consumption setting 1 Stopping internal circuit end timing designated (Refer to “List of Settings” item.)		0000h	00h	S		
		1								
		2								
		3								
		4								
		5								
		6								
		7								
A7h	35A7h	0		MSB			00h			
		1								
		2								
		3								
		4								
		5								
		6								
		7								

Address		bit	Register Name	Description	Default value after reset				Reflection timing		
4-wire	I ² C				By register		By address				
A8h	35A8h	0	HSST_START2 [11:0]	LSB	29Ah ^{*2}	FFFh ^{*5}	9Ah ^{*2}	FFh ^{*5}	S		
		1									
		2									
		3									
		4									
		5									
		6									
A9h	35A9h	7	WCLP	Horizontal blank low power consumption setting 2 Stopping internal circuit start timing designated (Refer to “List of Settings” item.)	29Ah ^{*2}	FFFh ^{*5}	9Ah ^{*2}	FFh ^{*5}	S		
		0									
		1									
		2									
		3									
		4									
		5									
AAh	35AAh	6	HSST_EN2 [0]	Horizontal blank low power consumption setting 2 enable 0: Disable, 1: Enable	1 ^{*2}	0 ^{*5}	42h ^{*2}	0Fh ^{*5}	S		
		7		Fixed to 0	0						—
		0	HSST_END2 [15:0]	LSB	0000h	00h			00h	S	
1											
2											
3											
4											
5											
6											
ABh	35ABh	7	HSST_END2 [15:0]	Horizontal blank low power consumption setting 2 Stopping internal circuit end timing designated (Refer to “List of Settings” item.)	0000h	00h	00h	S			
		0									
		1									
		2									
		3									
		4									
		5									
6											
		7		MSB							

Address		bit	Register Name	Description	Default value after reset				Reflection timing
4-wire	I ² C				By register		By address		
ECh	35ECh	0	HSST_START3 [11:0]	LSB	29Ah ^{*2}	FFFh ^{*5}	9Ah ^{*2}	FFh ^{*5}	S
		1		Horizontal blank low power consumption setting 3 Stopping internal circuit start timing designated (Refer to "List of Settings" item.)					
		2							
		3							
		4							
		5							
		6							
		7							
EDh	35EDh	0	MSB	Horizontal blank low power consumption setting 3 enable 0: Disable, 1: Enable	1 ^{*2}	0 ^{*5}	12h ^{*2}	0Fh ^{*5}	S
		1							
		2							
		3							
		4	HSST_EN3 [0]	Fixed to 0	0	0	0	—	
		5							
		6							
		7							
EEh	35EEh	0	HSST_END3 [15:0]	LSB	0000h		00h	S	
		1		Horizontal blank low power consumption setting 3 Stopping internal circuit end timing designated (Refer to "List of Settings" item.)					
		2							
		3							
		4							
		5							
		6							
		7							
EFh	35EFh	0	MSB	Horizontal blank low power consumption setting 3 Stopping internal circuit end timing designated (Refer to "List of Settings" item.)	0000h		00h	S	
		1							
		2							
		3							
		4							
		5							
		6							
		7							

^{*2} IMX530 series and IMX540 series.

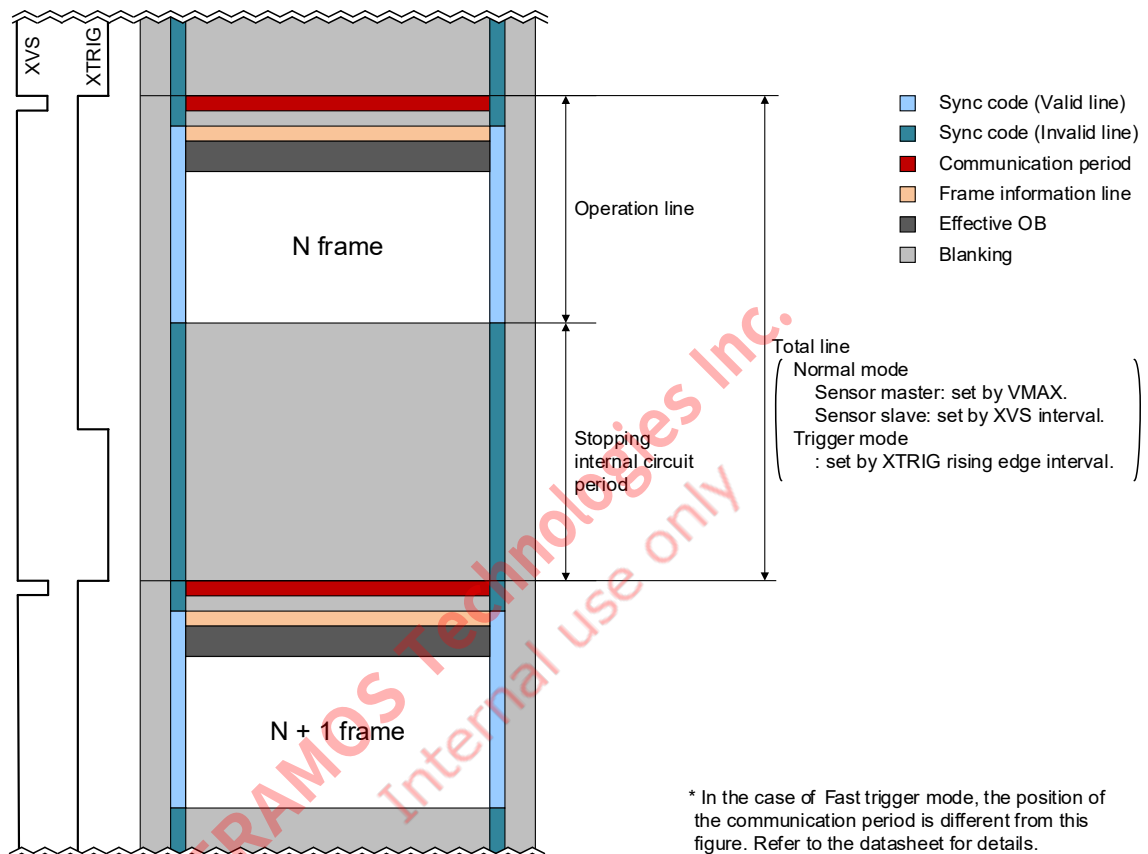
^{*5} IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

5. About Vertical Blanking Low Power Consumption Setting

5.1. Overview

This function stops the sensor internal circuit during the vertical line period when sensor readout is not operating. In this sensor, this function is always ON.
The image of SLVS output is shown below.

Ex.) In the case of SLVS output



Vertical Blanking Low Power Consumption Setting Diagram

5.2. List of Settings

Operation Mode	VSST_EN
All-Pixels	0Fh
ROI	
Vertical / Horizontal 1/2 Subsampling	
2 × 2 Vertical Binning ^{*1}	

^{*1} The product with a color filter is excluded

6. About Horizontal Blanking Low Power Consumption Setting

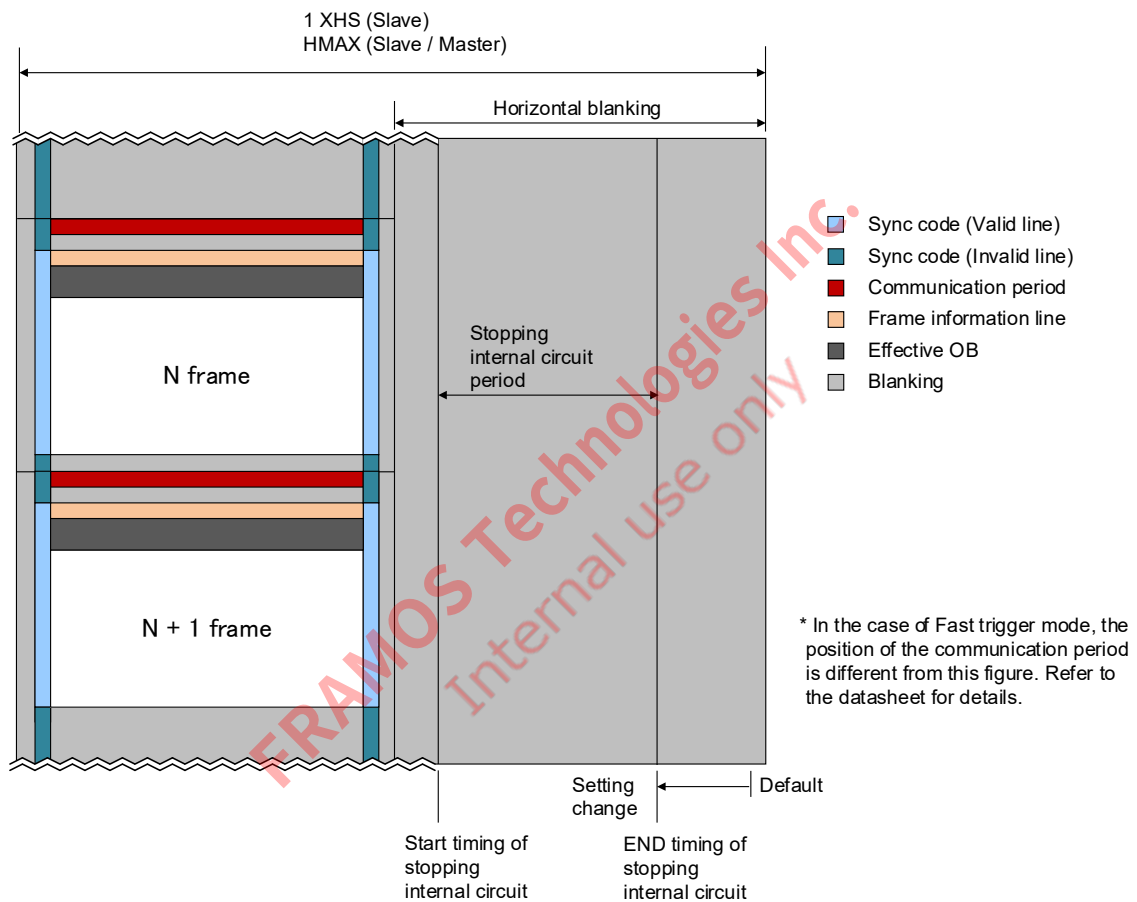
6.1. Overview

This function stops the sensor internal circuit during the horizontal blanking period when the sensor readout is not operating. In this sensor, this function is always ON. Hence, be sure to set HMAX [15:0] even in sensor slave mode (including Sequential trigger mode). The setting value of HMAX [15:0] can be calculated using the following.

$$\text{HMAX [15:0]} = \text{"Number of INCK in 1H"} \times 74.25 / \text{INCK Frequency}$$

The image of SLVS output is shown below.

Ex.) In the case of SLVS output



Horizontal Blanking Low Power Consumption Setting Diagram

6.2. List of Settings

Settings of IMX530 series and IMX540 series

Operation Mode	HSST_START1 HSST_START2 HSST_START3 setting [DEC]	HSST_END1 HSST_END2 HSST_END3 setting [DEC]
All-Pixels	12bit: 296h (662) 10bit: 295h (661) 8bit: 297h (663)	0000h
ROI		
Vertical / Horizontal 1/2 Subsampling		
2 × 2 Vertical Binning ^{*1}		

^{*1} The product with a color filter is excluded

Settings of IMX535 series, IMX545 series and IMX565 series

Operation Mode	HSST_START1 HSST_START2 HSST_START3 setting [DEC]	HSST_END1 HSST_END2 HSST_END3 setting [DEC]
All-Pixels	12bit, 8bit: 208h (520) 10bit: 21Ch (540)	0000h
ROI		
Vertical / Horizontal 1/2 Subsampling		
2 × 2 Vertical Binning ^{*1}		

^{*1} The product with a color filter is excluded

Settings of IMX548 series and IMX568 series

Operation Mode	HSST_START1 HSST_START2 HSST_START3 setting [DEC]	HSST_END1 HSST_END2 HSST_END3 setting [DEC]
All-Pixels	12bit: 230h (560) 10bit: 2D0h (720) 8bit: 262h (610)	0000h
ROI		
Vertical / Horizontal 1/2 Subsampling		
2 × 2 Vertical Binning ^{*1}		

^{*1} The product with a color filter is excluded

Settings of all series

Operation Mode	HSST_EN1 HSST_EN2 HSST_EN3 setting
All-Pixels	1
ROI	
Vertical / Horizontal 1/2 Subsampling	
2 × 2 Vertical Binning ^{*1}	

^{*1} The product with a color filter is excluded

7. Note

- Normal mode (XVS / XHS Sync), Sequential trigger mode and Fast Trigger mode are supported.
- Sensor master mode and sensor slave mode are supported.
When using with the sensor slave mode, set the "Number of INCK in 1H" $\times$ 74.25 / INCK Frequency to HMAX [15:0].
- This function can be used with the following functions.
 - Multi Frame Set Output mode (2 / 4 frame)
 - Multi Exposure Trigger mode
 - Multi Frame ROI (Multi Exposure + ROI) mode
 - Dual ADC (without combination / with built-in combination) ^{*1}
 - Dual Trigger
 - Self Trigger
 - Short interval shutter function

^{*1} IMX540 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series are excluded.

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8. Trouble case

The following troubles have been reported in relation to this function, and the causes and countermeasures are described below.

[Phenomenon]

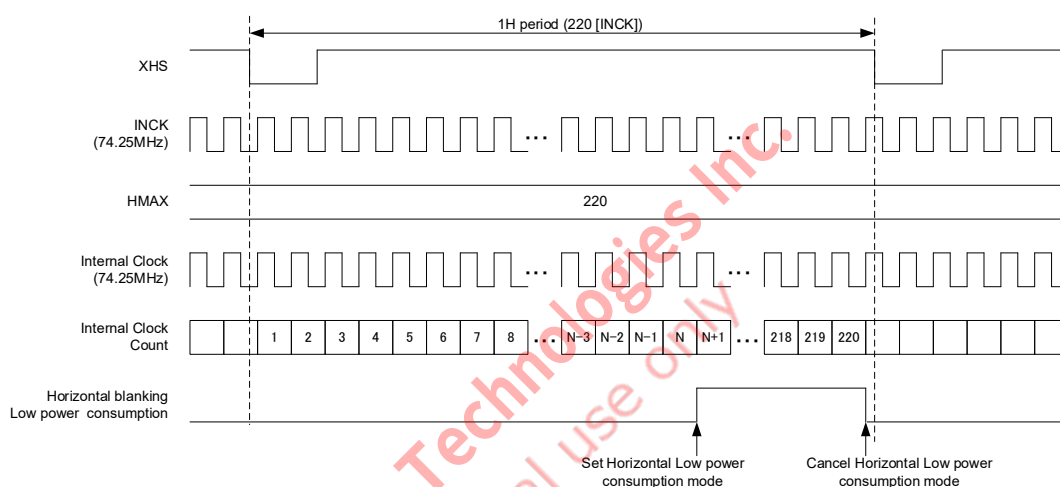
- There is no output image.
- The output image is broken.

[Cause]

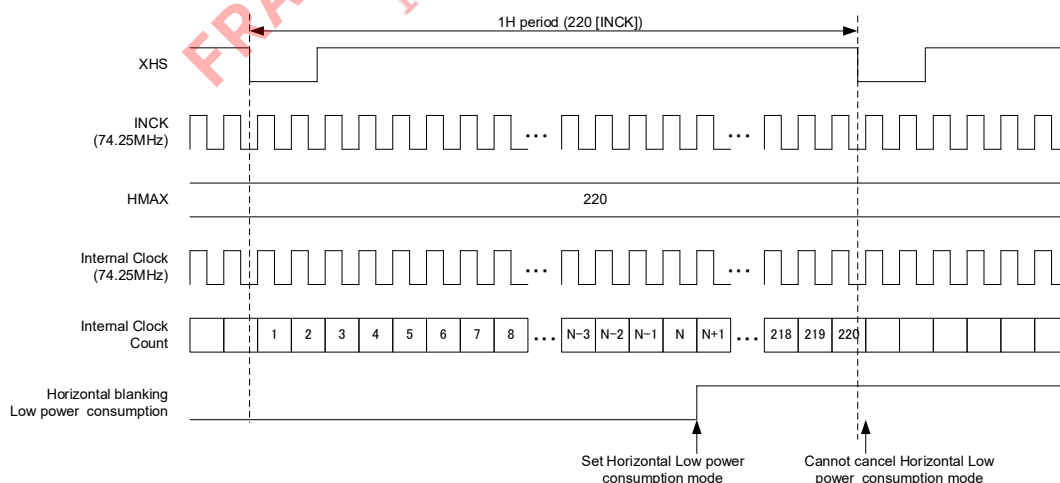
Horizontal blank Low power consumption is controlled by the internal clock (74.25MHz) generated from INCK and HMAX.

If the clock sources (oscillator) that generate INCK and XHS are different, the number of INCKs in 1H may be ± 1 depending on the phase between INCK and XHS due to a frequency error.

It occurs when the Horizontal blanking low power consumption function is not canceled.



Normal case

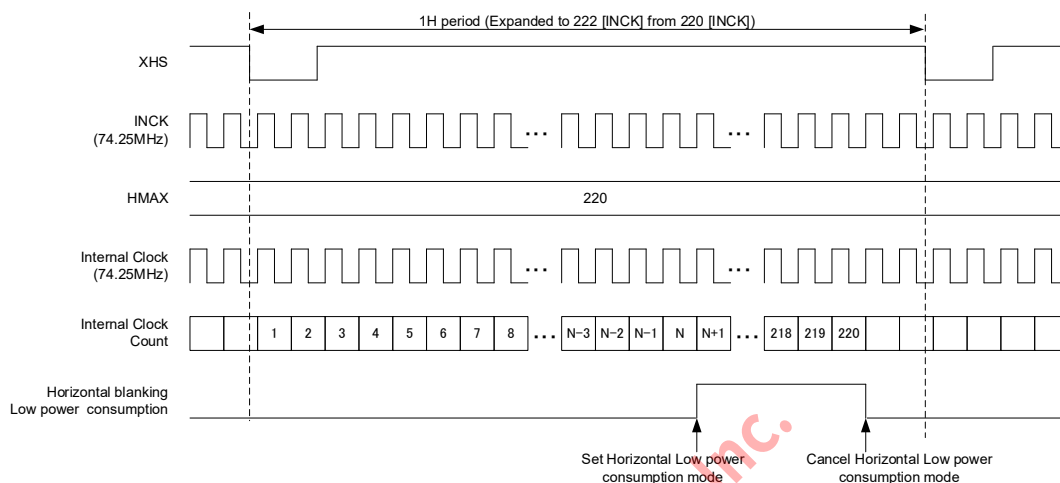


When the number of INCKs in 1H is short

[Countermeasure]

Use one of the following.

- Expand the XHS input period (the number of INCKs in 1H).
Even if there is a frequency error in the clock that generates XHS and INCK, by increasing the number of INCKs in 1H by +2, the low power consumption function in the horizontal direction will always be canceled.



- Disable the horizontal blank low power consumption function.
Set 0 in the registers below.

Chip ID	Address	bit	Register Name	Setting
07h	A5h	[4]	HSST_EN1 [0]	Horizontal blank low power consumption setting 1 enable 0: Disable 1: Enable
	A9h	[6]	HSST_EN2 [0]	Horizontal blank low power consumption setting 2 enable 0: Disable 1: Enable
	EDh	[4]	HSST_EN3 [0]	Horizontal blank low power consumption setting 3 enable 0: Disable 1: Enable

Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 4 / 26	–	First Edition
Rev.0.2	2019 / 6 / 20	–	Correction: Document name IMX530_AppNote_Driving_Low_Power_Consumption → IMX530_AppNote_Low_Power_Consumption
		1, 3	Correction: The function name of driving low power consumption at longtime exposure
		1	Correction: Explanation of a longtime exposure of description
		10	Correction: The function name of twice exposure
Rev.1.0	2019 / 12 / 9	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added: Chapter "Product type" Delete: Notes on using horizontal blanking low power consumption setting Added: Low power consumption settings are always ON. Correction: 2 × 2 FD binning mode Added: "The product with a color filter is excluded." Added: SLVS output (2 ch) in the support interface
		3	Correction: The explanation of the value set for HMAX has been changed from text to a formula.
		–	Delete: Registers corresponding to Chip ID = 07h
		4	Added: Vertical blanking low power consumption setting is always ON. Correction: Figure "Vertical Blanking Low Power Consumption Setting Diagram" Figure that considered only the Normal mode → figure that also considered the Trigger modes
		5	Added: Horizontal blanking low power consumption setting is always ON. Added: HMAX must be set even in Sensor slave mode. Added: The formula to calculate the value to be set in HMAX Correction: Figure "Horizontal Blanking Low Power Consumption Setting Diagram" Delete: "Setting Value"
		–	Delete: Section "Setting Value" of the horizontal blanking low power consumption setting
		6	Correction: The explanation of the value set for HMAX has been changed from text to a formula. Delete: Note on ON / OFF of low power consumption settings
Rev.2.0	2020 / 12 / 17	–	Added: IMX535 series / IMX545 series / IMX565 series
		2	Added: CSI-2 serial data output
		4	Added: Default Value after reset of IMX535 series, IMX545 series, IMX565 series

Version	Date	Page	Remarks
Rev.3.0	2021 / 2 / 18	1, 2	Added: IMX548 series / IMX568 series
		4	Added: Default Value after reset of IMX548 series, IMX568 series
		5, 6, 7	Added: Registers corresponding to Chip ID = 07h
		8	Added: List of Settings
		10	Added: List of Settings
		11	Added: Explanatory note for Dual ADC
Rev.4.0	2022 / 1 / 27	10	Added: HSST_EN1 to HSST_EN3 to List of Settings
		12	Added: Chapter "Trouble case"
Rev.5.0	2023 / 1 / 5	6	Correction: description of WCLP register

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